

# Linear vs Nonlinear Contrastive Encoders

## Main question

**When  $Y$  is a known deterministic function of  $X$  plus noise, does the nonlinear encoder move toward the ideal PVE reference or mostly memorize training pairs?**

This deck tests whether contrastive encoders recover deterministic signal or memorize paired samples, using noise controls, SNR sweeps, overfitting diagnostics, and probe readouts.

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# Three checks, one training protocol

The updated deck separates pure memorization from deterministic signal recovery and probe-readout interpretability.

## Noise only

X and Y are independent Gaussian noise. Any train-only alignment is memorization.

## Deterministic linear

$Y = f(X) + \epsilon$  with  $f(X)=X$ . Target SNR controls the clean signal versus noise.

## Deterministic cubic

$Y = f(X) + \epsilon$  with  $f(X)=X^3$ . This stresses nonlinear recovery.

## SNR/PVE reference

The red reference line is the ideal clean/noisy cosine separation implied by SNR; the log x-axis uses the same SNR values.

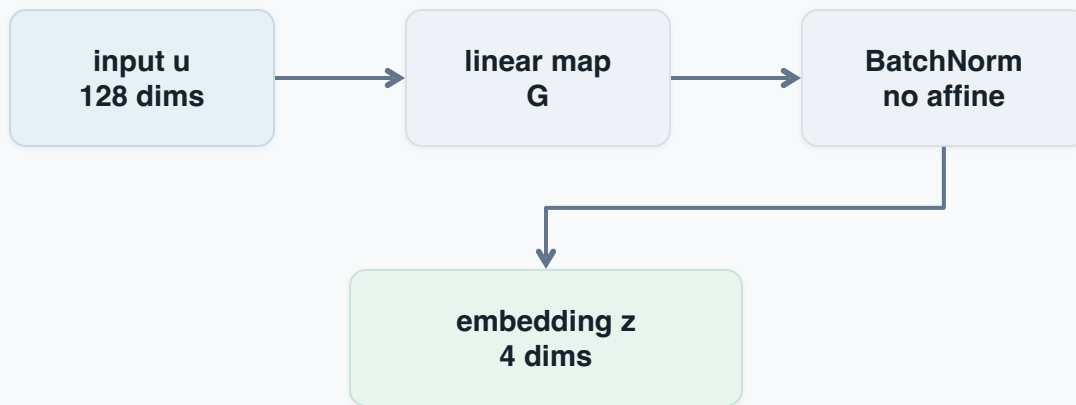
## Probe experiment

Freeze the trained encoder, fit ridge readouts from branch embeddings to the clean target, and sweep one raw coordinate.

# Matched encoders with a controlled residual branch

The architecture comparison stays fixed while the deterministic relationship and SNR vary.

## Pure linear encoder



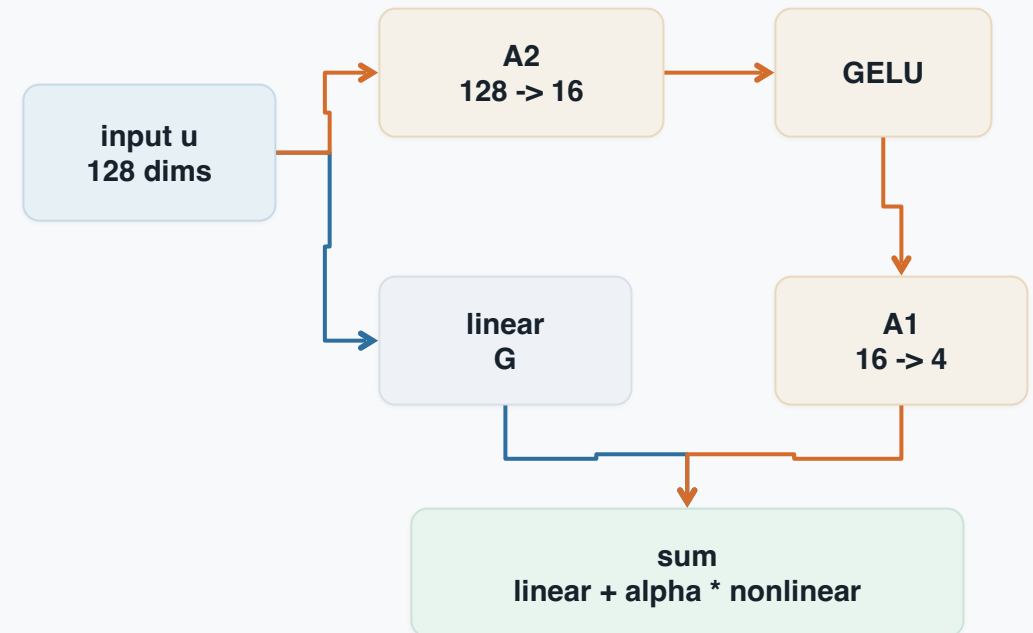
$$g(u) = \text{Normalize}(G u)$$

shared X/Y family

embedding dim = 4

hidden dim = 16

## Residual nonlinear encoder



$$g(u) = \text{Normalize}(G u) + \alpha * \text{Normalize}(A_1 \text{ GELU}(A_2 u + b))$$

# Paper contrastive loss with optional penalties

All model families use the same objective; differences come from architecture and regularization.

## Similarity scores

For a batch of paired embeddings:

$$s_{i,j} = \mathbf{z}_{x,i}^T \mathbf{z}_{y,j}$$

$$\text{maximize } L = \sum_i s_{ii} - \sum_i \log \sum_j \exp(s_{i,j}) - \sum_j \log \sum_i \exp(s_{i,j})$$

training minimizes  $-L / n$  plus any active regularization terms

## L1 penalties

Linear:  $\|\mathbf{G}_x\|_1 + \|\mathbf{G}_y\|_1$

Nonlinear:  $\|\mathbf{A}\|_1$  terms when enabled.

## L2 penalties

Linear:  $\|\mathbf{G}_x\|_2^2 + \|\mathbf{G}_y\|_2^2$

Nonlinear:  $\|\mathbf{A}_1\|_2^2 + \|\mathbf{A}_2\|_2^2$ .

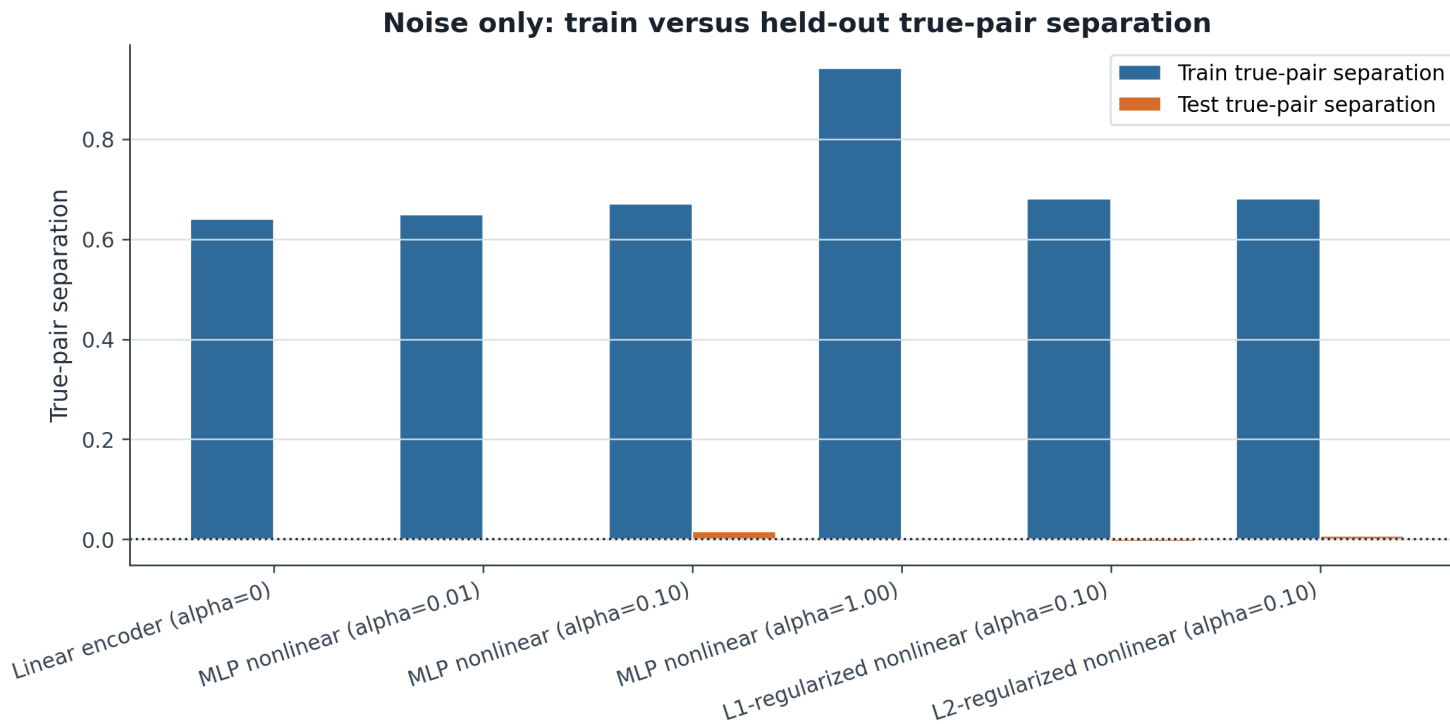
## Output penalty

Available to keep the actual nonlinear correction small unless needed.

## RESULT

# Noise-only: train vs held-out separation

High train separation with low held-out separation indicates memorization rather than signal.



## Interpretation

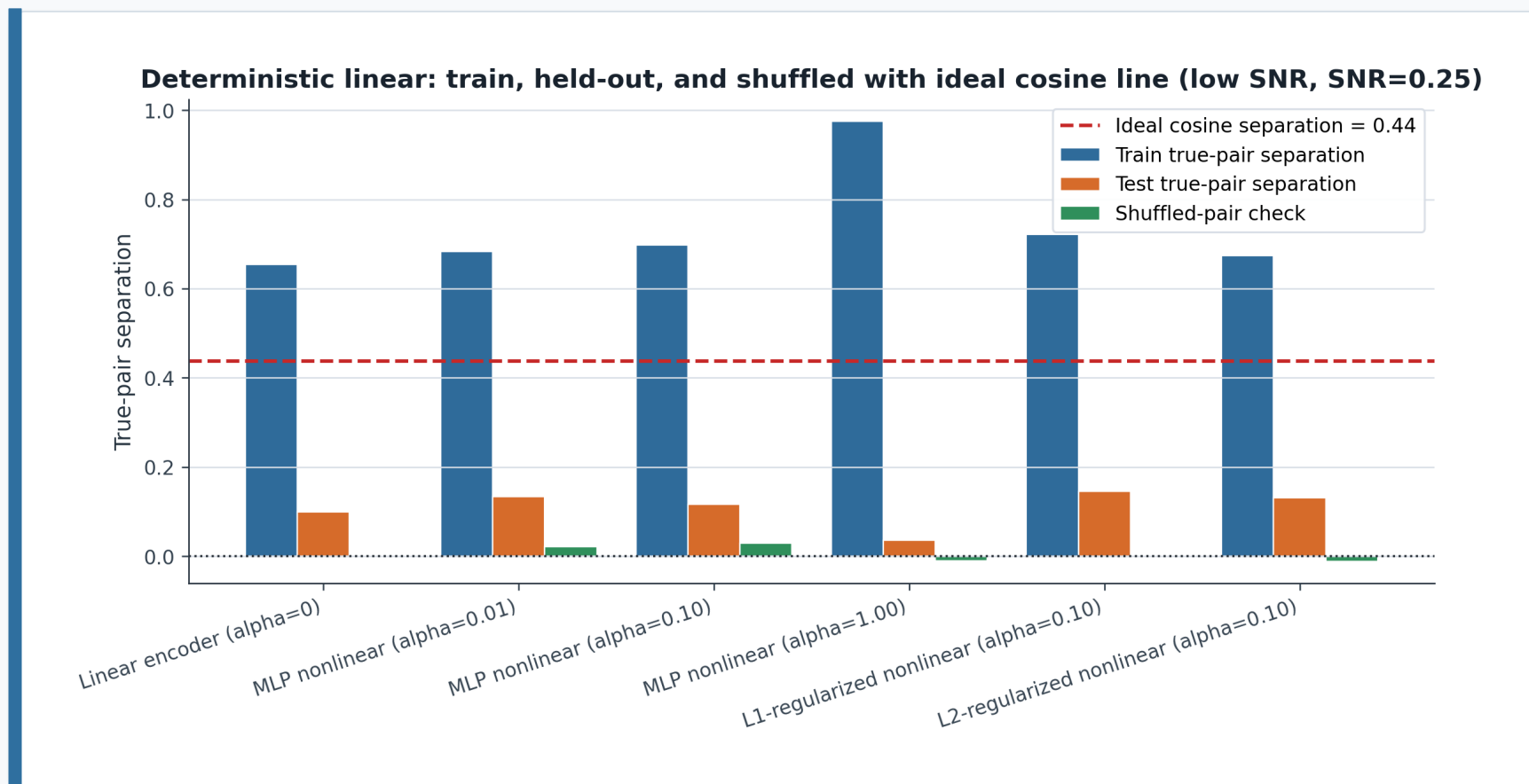
In noise-only data, X and Y have no real relationship. Training alignment can still grow because the model can memorize paired training samples.

## Key takeaway

Held-out performance remains the sanity check: high training alignment alone is not evidence of real learning.

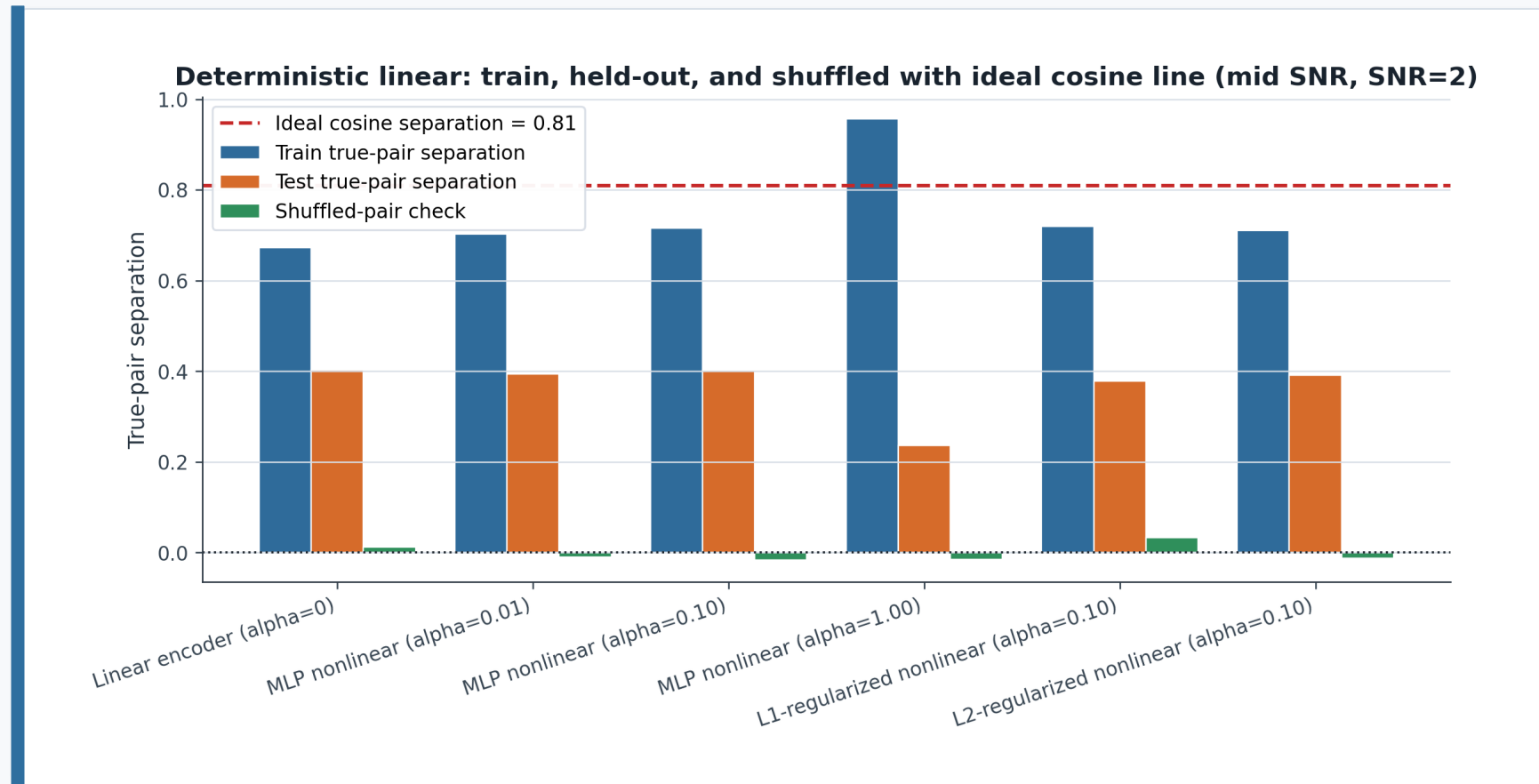
## Deterministic linear: train, held-out, and shuffled (low SNR, SNR=0.25)

Low-SNR snapshot with the ideal cosine reference line.



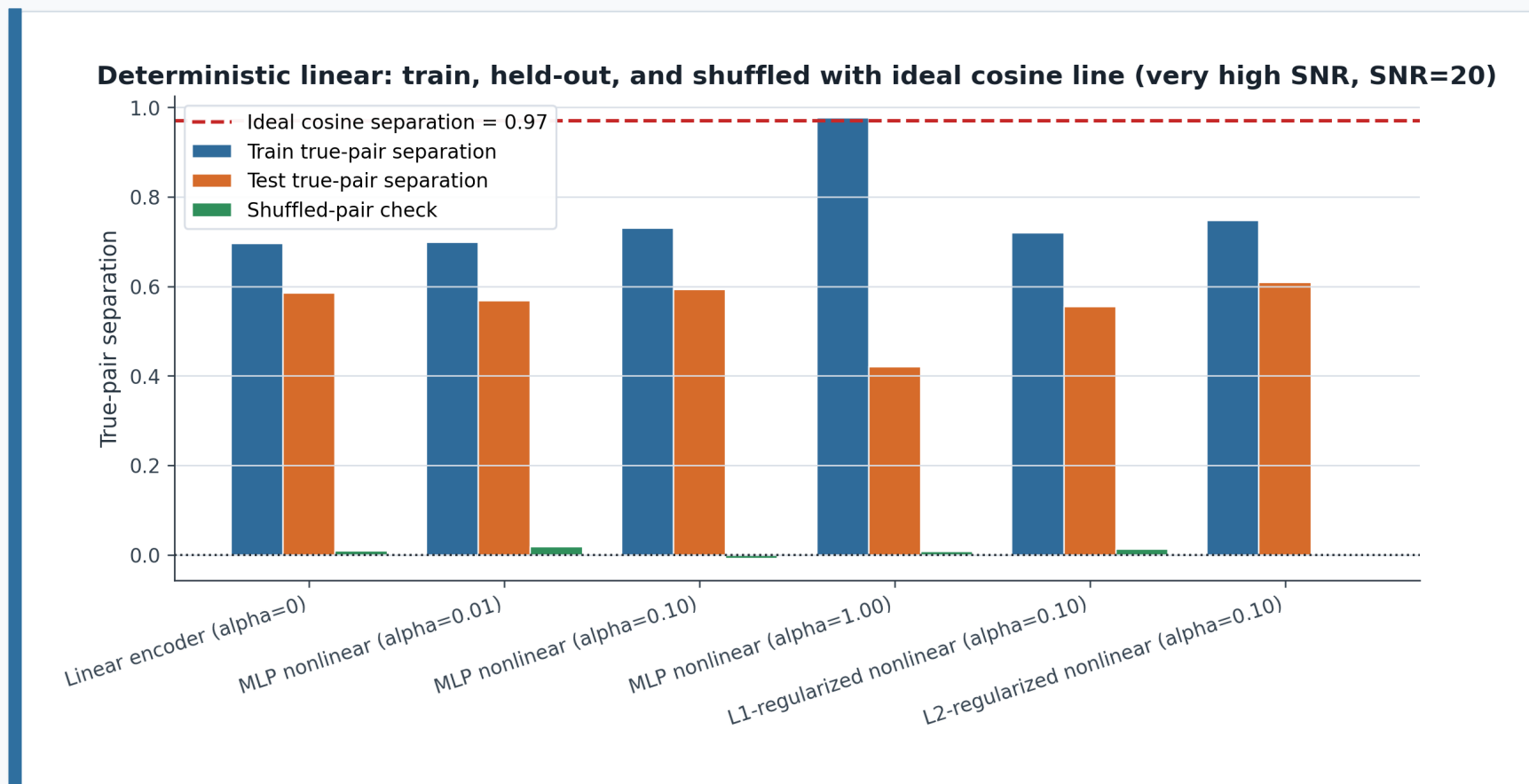
# Deterministic linear: train, held-out, and shuffled (mid SNR, SNR=2)

Mid-SNR snapshot with the ideal cosine reference line.



## Deterministic linear: train, held-out, and shuffled (very high SNR, SNR=20)

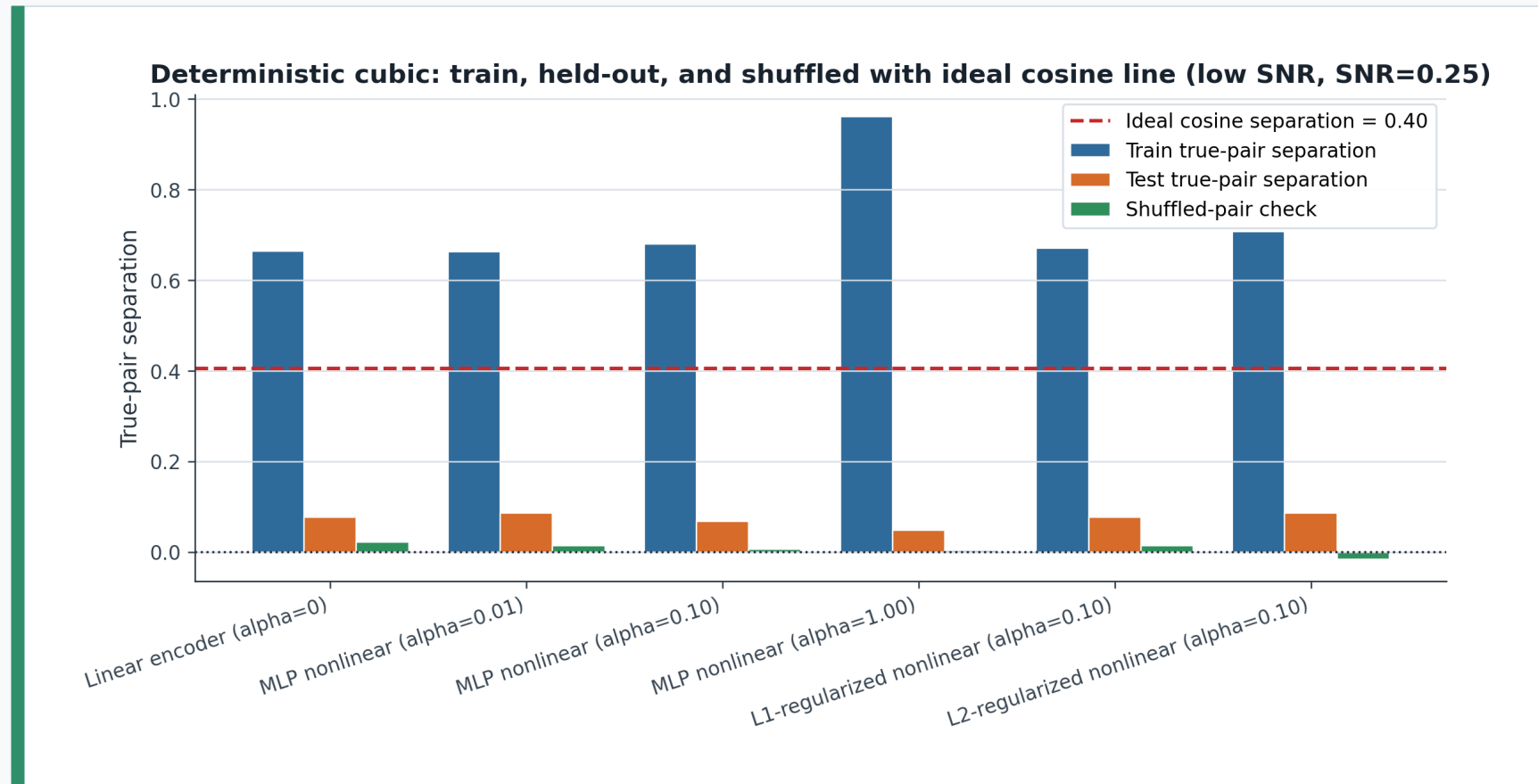
Very-high—SNR snapshot with the ideal cosine reference line.





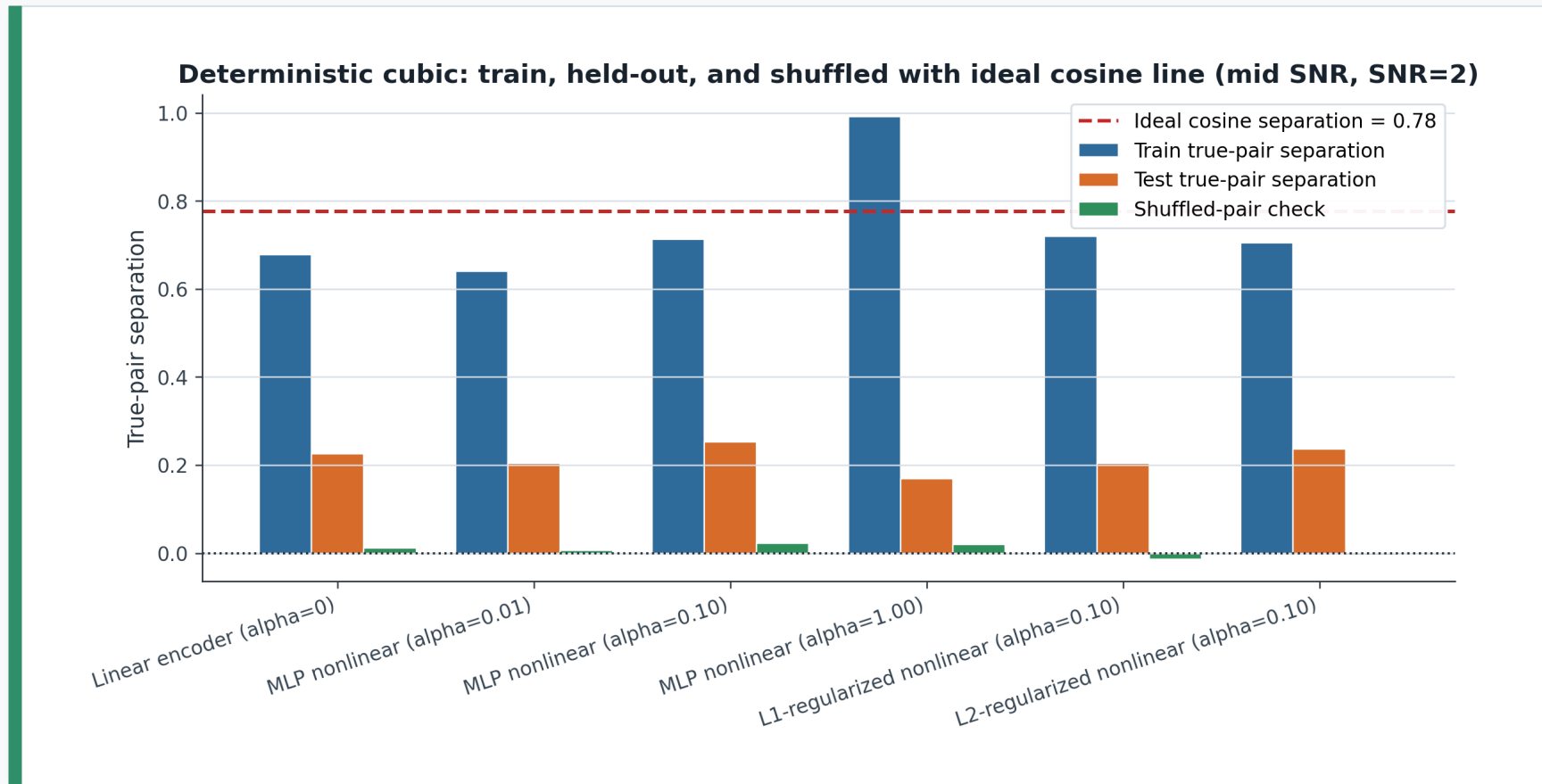
# Deterministic cubic: train, held-out, and shuffled (low SNR, SNR=0.25)

Low-SNR snapshot with the ideal cosine reference line.



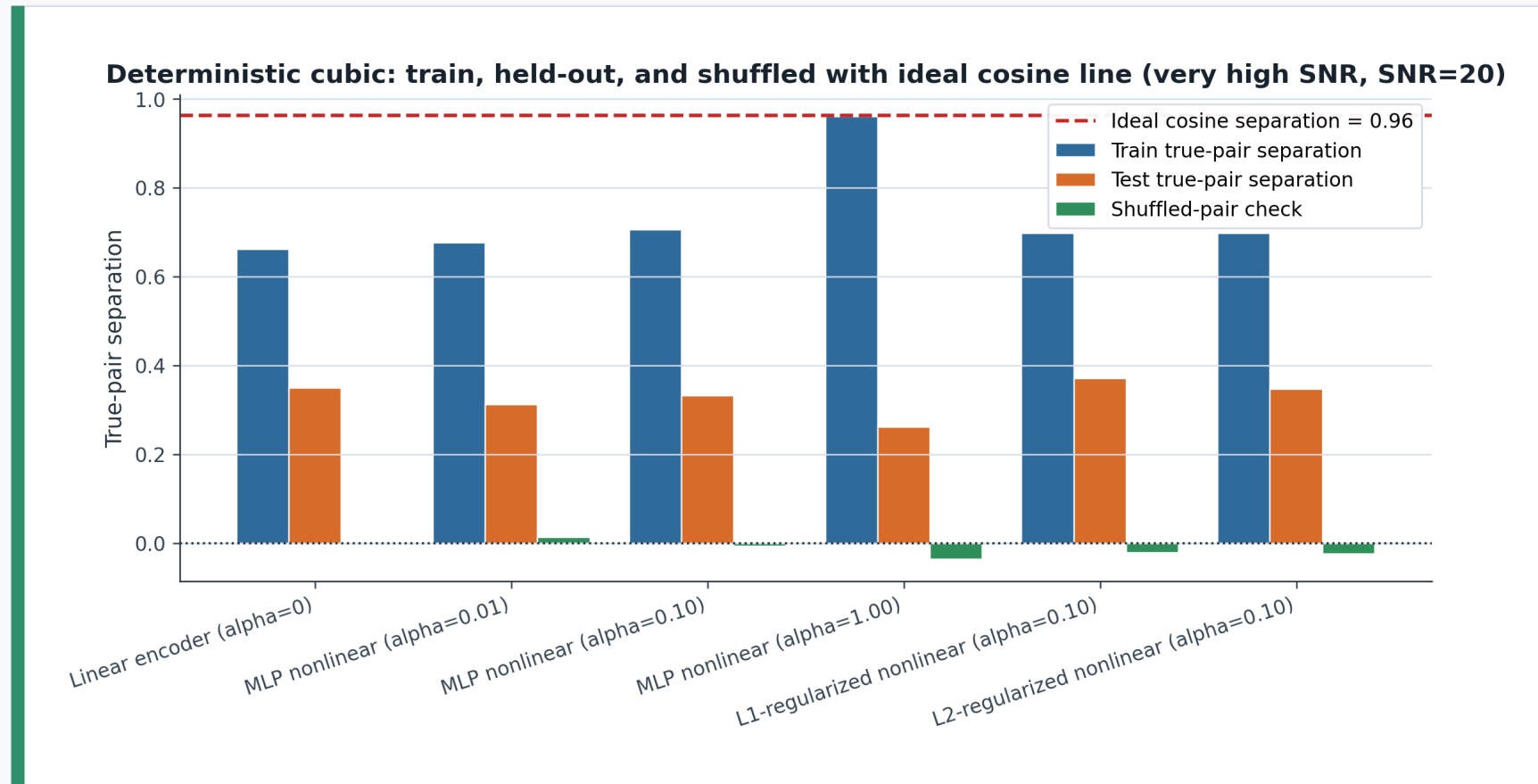
# Deterministic cubic: train, held-out, and shuffled (mid SNR, SNR=2)

Mid-SNR snapshot with the ideal cosine reference line.



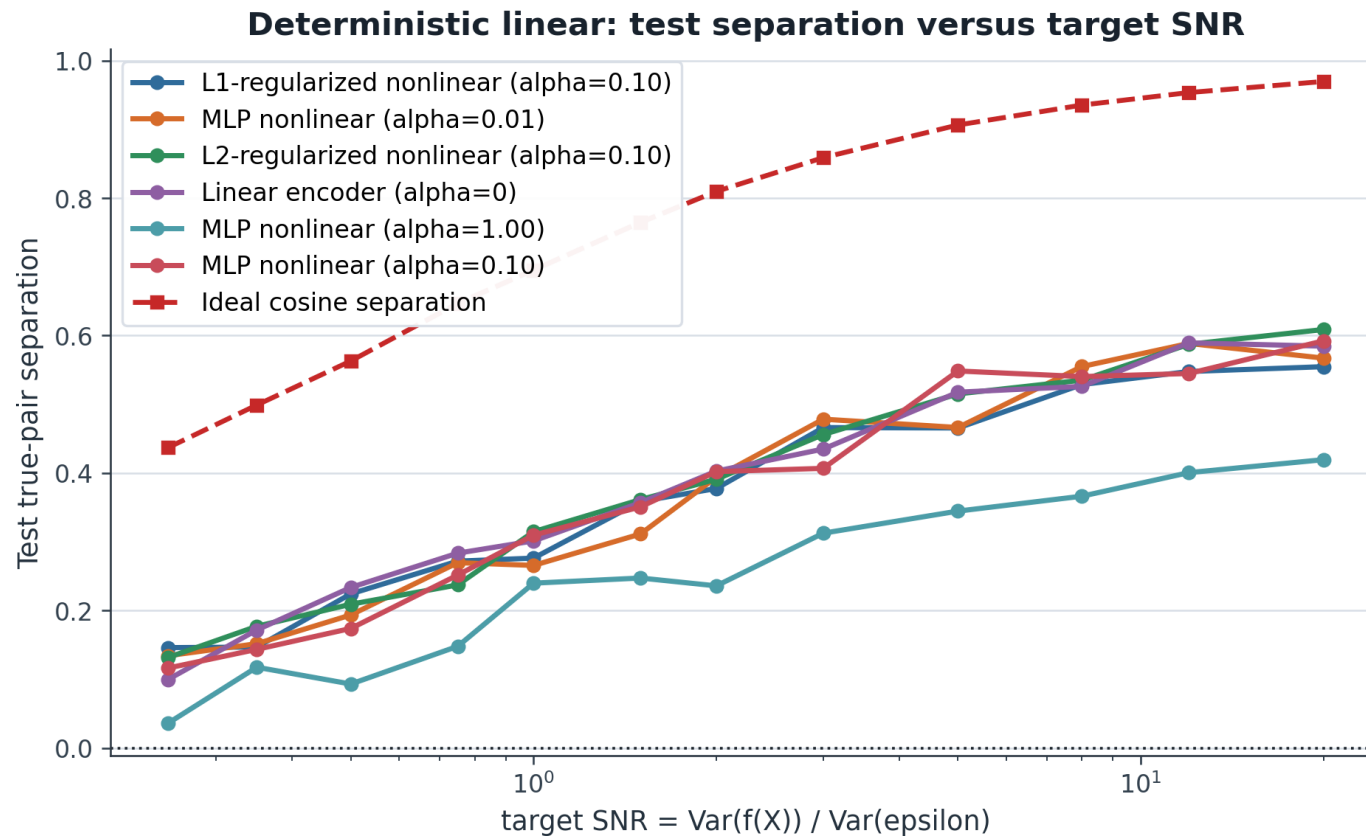
# Deterministic cubic: train, held-out, and shuffled (very high SNR, SNR=20)

Very-high-SNR snapshot with the ideal cosine reference line.



# Deterministic linear: test separation vs SNR

The many-point log-SNR sweep includes the red ideal PVE/cosine reference line.



## What to read

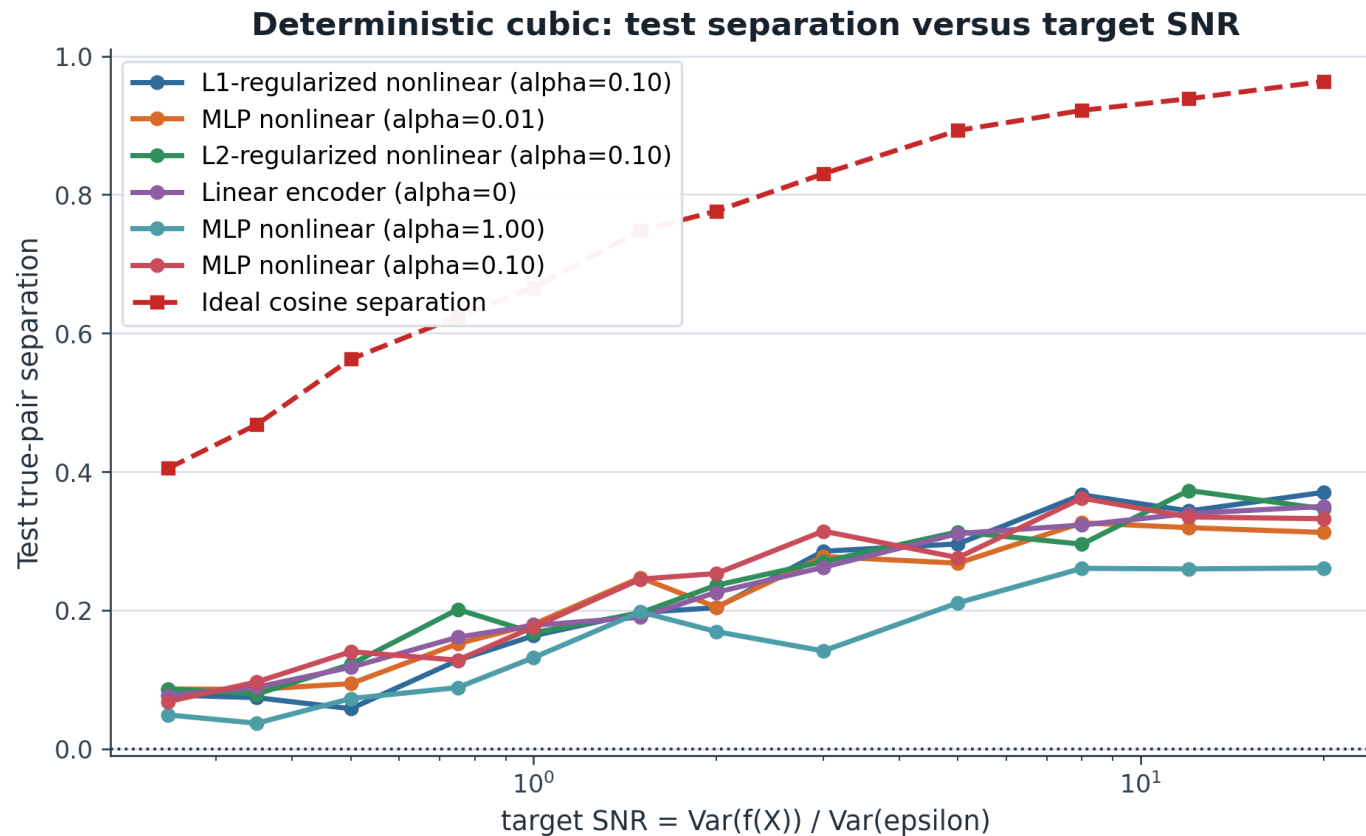
Test separation improves as SNR rises, but all learned encoders remain well below the red ideal reference across the sweep.

## Key takeaway

The linear deterministic task is not solved just by increasing SNR; model-side recovery remains limited.

# Deterministic cubic: test separation vs SNR

The cubic relationship is harder; the same red ideal reference provides a clean upper benchmark.



## What to read

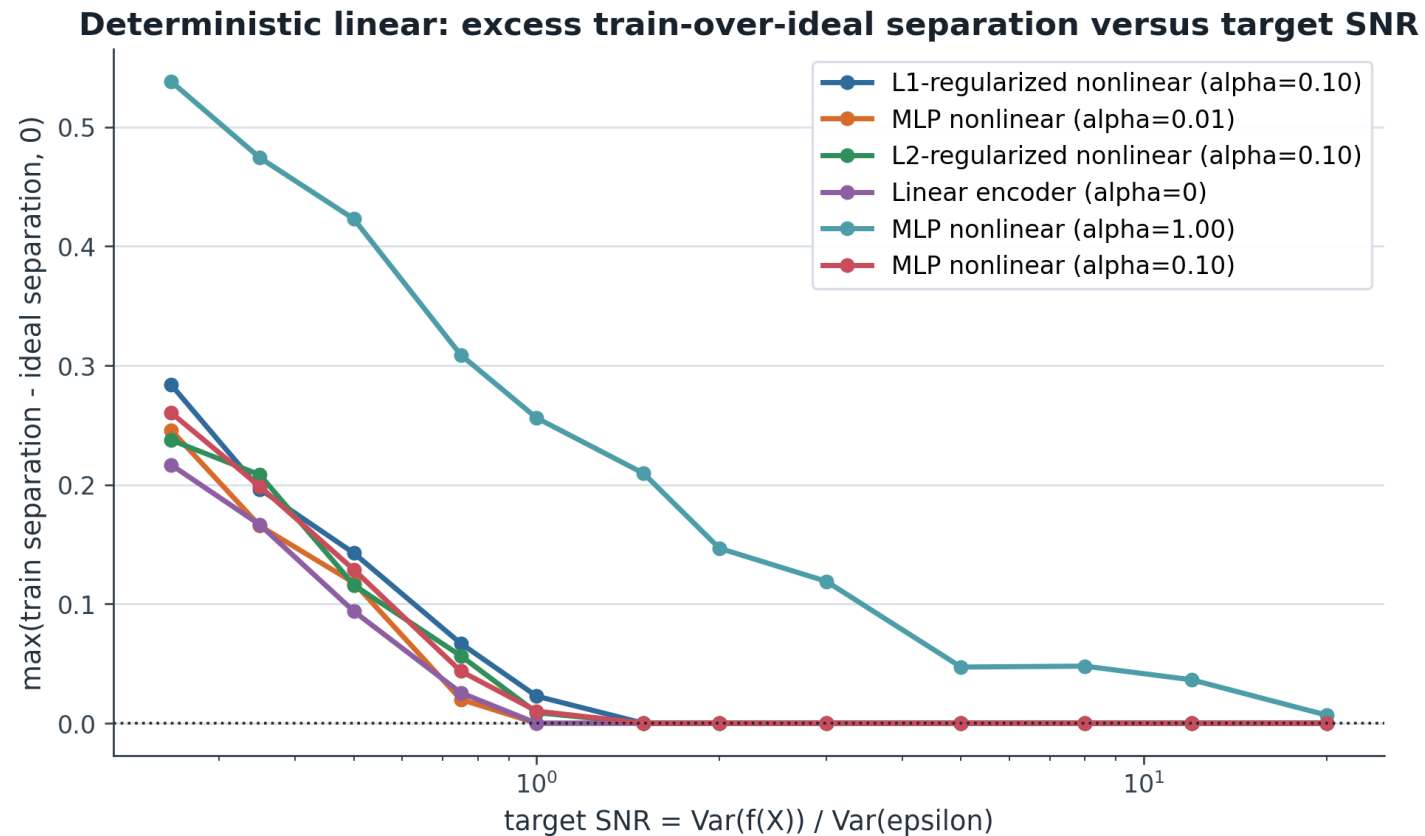
The cubic test curves rise with SNR, but the gap to the ideal reference is larger than in the linear case.

## Key takeaway

Controlled nonlinear corrections help somewhat, but they do not close the deterministic recovery gap.

# Linear overfitting: train score above the ideal

The log-axis excess train-over-ideal plot quantifies train-side memorization beyond the PVE reference.



## Overfitting measure

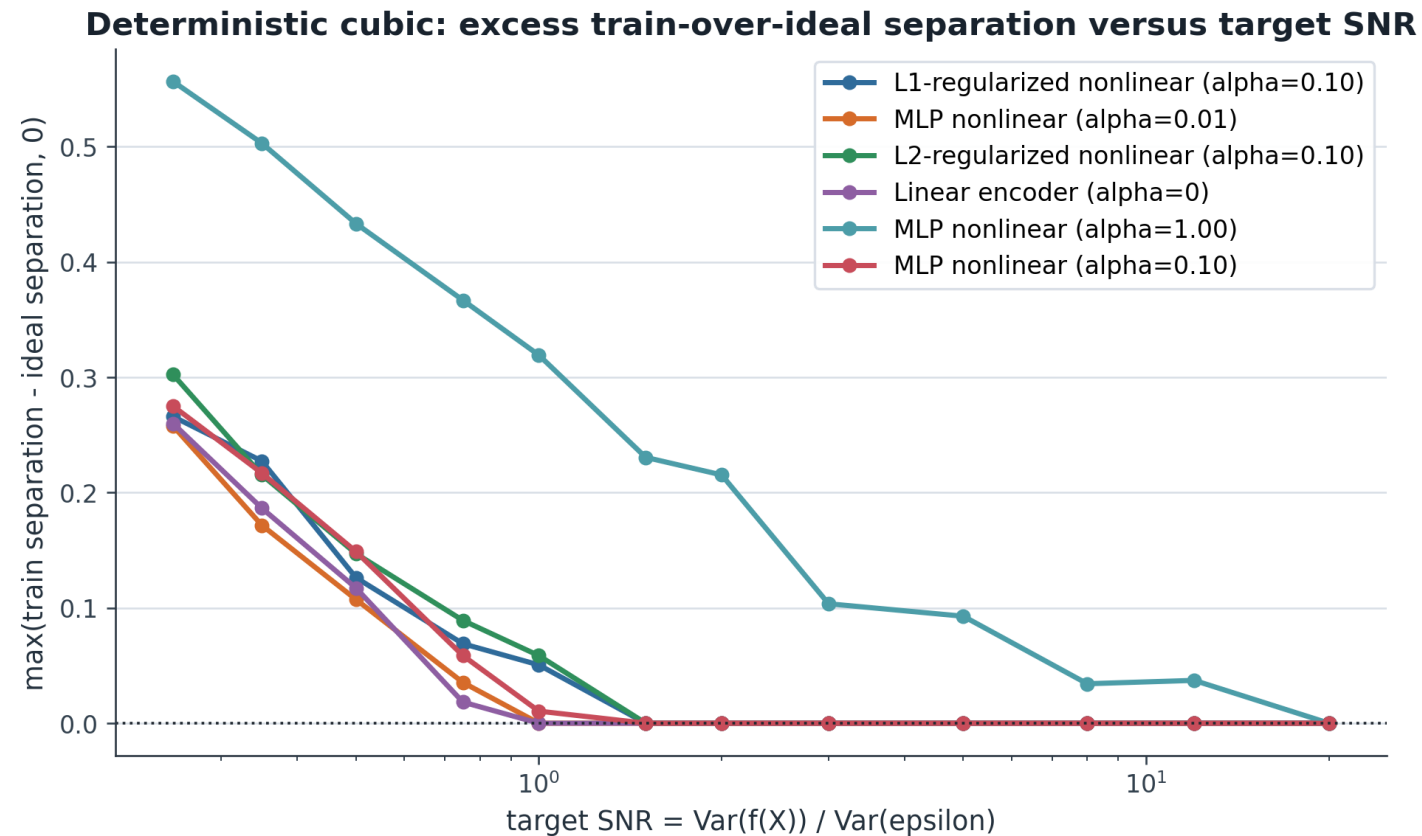
This plot shows  $\max(\text{train separation} - \text{ideal separation}, 0)$ . Low SNR leaves room for models to fit training pairs beyond the clean/noisy reference.

## Key takeaway

The  $\alpha=1$  nonlinear model shows the largest excess, making it the clearest overfitting warning.

# Cubic overfitting: train score above the ideal

The same log-axis diagnostic shows how quickly excess train alignment disappears as SNR improves.



## Overfitting measure

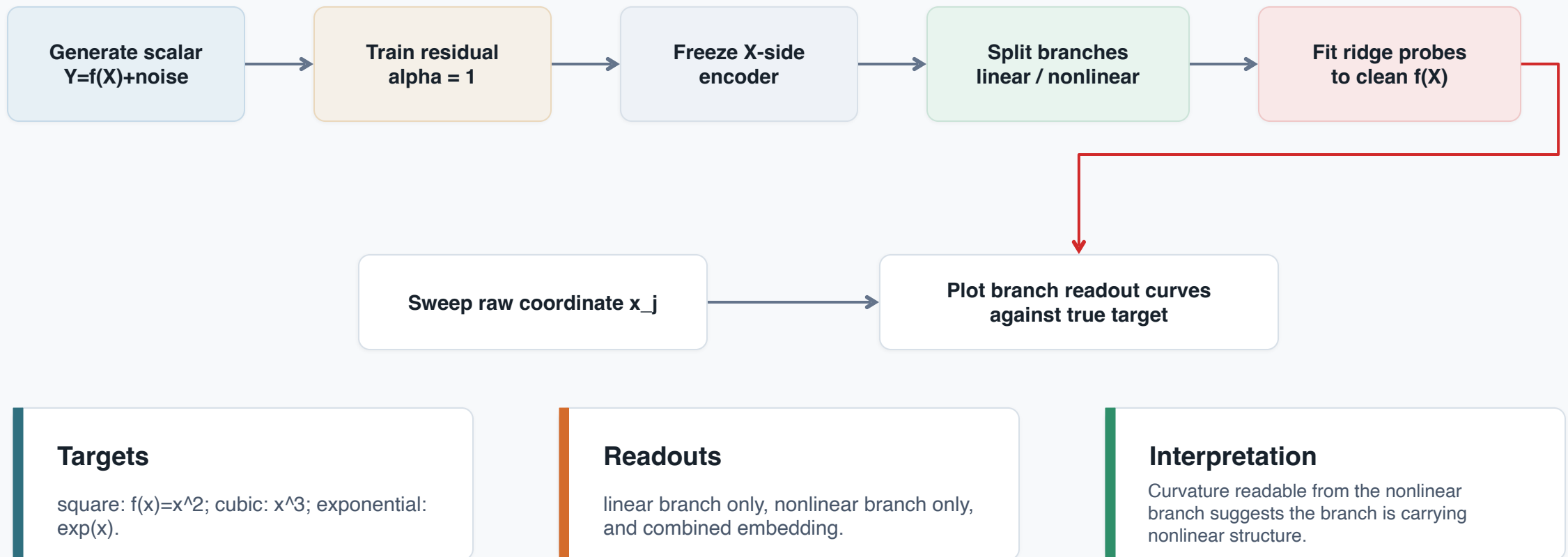
Cubic excess train-over-ideal is largest in the low-SNR regime and shrinks toward zero for most models as signal becomes easier.

## Key takeaway

Again, the alpha=1 nonlinear model stays suspicious longer than controlled or regularized settings.

# Pipeline for reading out learned nonlinear information

The probe is fitted after training, so it measures what the frozen encoder made linearly readable.

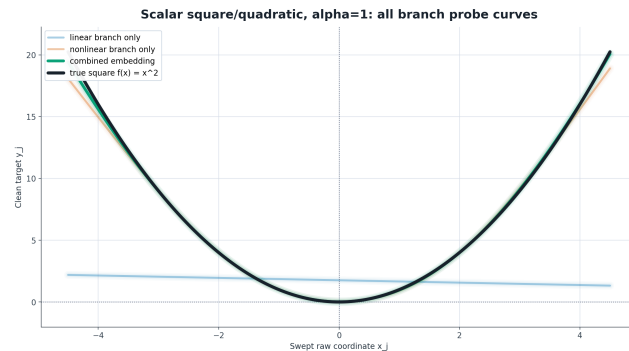




# Probe results across square, cubic, and exponential targets

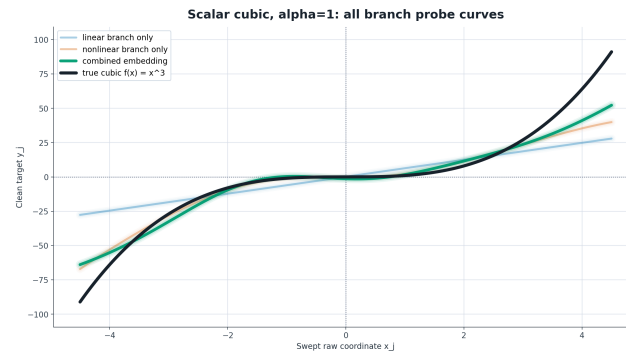
Branch-probe outputs show what is linearly readable from learned embeddings across nonlinear target functions.

## Square branch readout



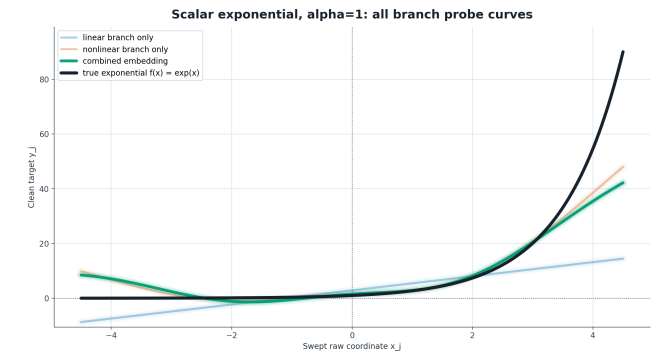
For  $f(x)=x^2$ , the combined embedding closely follows the target curve; the nonlinear branch carries the curvature.

## Cubic branch readout



The combined embedding follows the cubic shape better than either branch alone over part of the sweep.

## Exponential branch readout

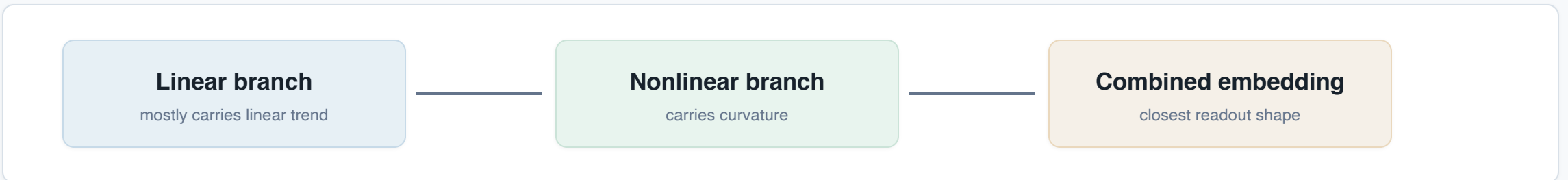


The combined and nonlinear-branch readouts pick up rising curvature, but still under-shoot the exponential tail.

Note: the scalar branch-probe notebook saves square/quadratic, cubic, and exponential branch-story plots, so all three panels use the same alpha=1 scalar readout format.

# Probe finding: nonlinear branch recovers nonlinear structure

The branch probes suggest that curvature is made linearly readable by the learned nonlinear component.



## What the plots show

For square and cubic targets, the nonlinear branch and combined embedding follow the curved target better than the linear branch.

## Interpretation

The residual branch is not just extra capacity: a ridge probe can read out nonlinear coordinate structure from the frozen embedding.

## Caution

Recovery is partial. The exponential target still under-shoots in the tail, so the probe supports nonlinear recovery, not perfect recovery.

**Takeaway:** the probe results give positive evidence that the nonlinear encoder is recovering the nonlinear part of the signal, while the SNR/PVE slides quantify how far that recovery remains from the ideal reference.

# Findings and next steps

## Main Finding

Nonlinear encoders learn real structure, but not perfectly.

## Evidence

Noise-only fails on held-out data.

Linear target approaches ideal behavior.

Nonlinear branch recovers curved structure in probe tests.

## Next steps

Compare square, cubic, and exponential probe results more systematically, test more random seeds, and check whether stronger nonlinear models reduce the gap from the ideal reference.